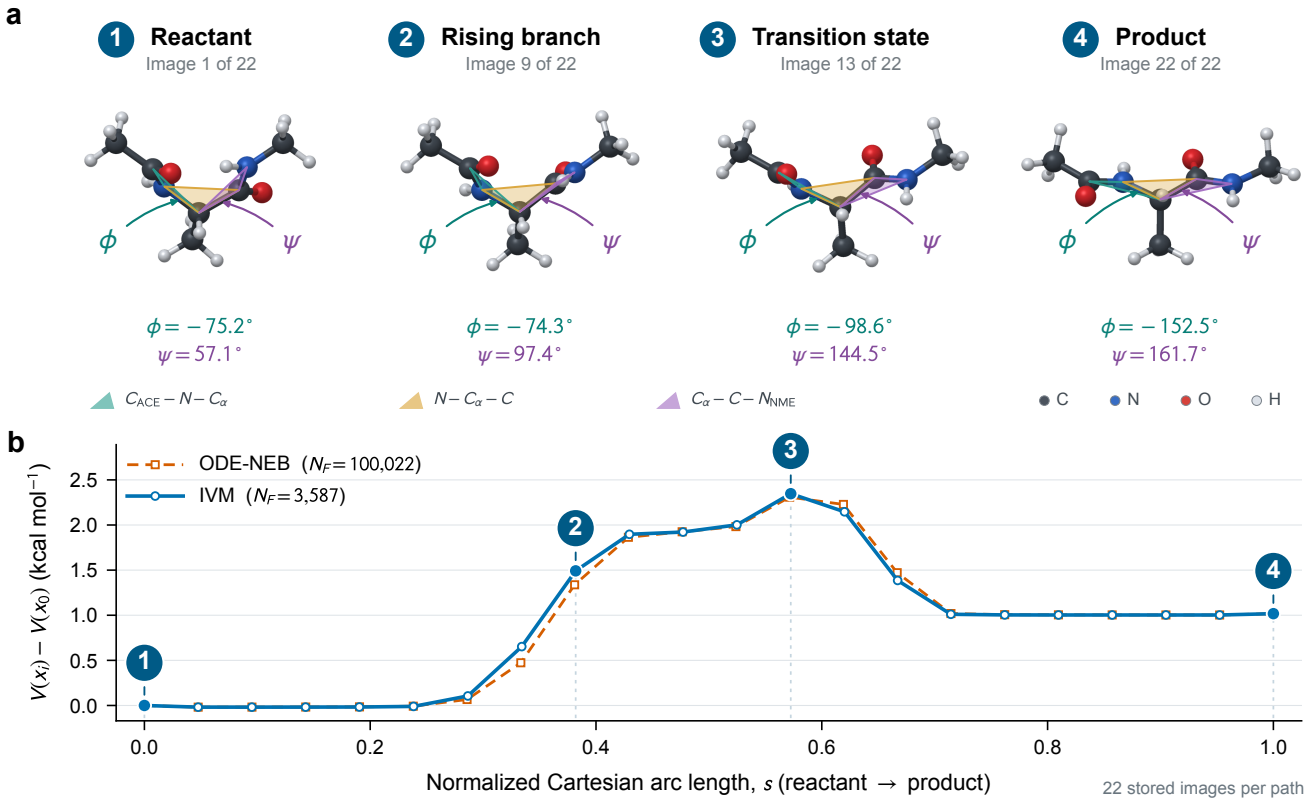

a
1 Reactant
Image 1 of 22
2 Rising branch
Image 9 of 22
3 Transition state
Image 13 of 22
4 Product
Image 22 of 22
ϕ
ψ
ϕ = −75.2°
ψ = 57.1°
ϕ = −74.3°
ψ = 97.4°
ϕ = −98.6°
ψ = 144.5°
ϕ = −152.5°
ψ = 161.7°
C_ACE − N − C_α
N − C_α − C
C_α − C − N_NME
C
N
O
H
b
ODE-NEB (N_F = 100,022)
IVM (N_F = 3,587)
V(x_i) − V(x_0) (kcal mol⁻¹)
0.0
0.5
1.0
1.5
2.0
2.5
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Cartesian arc length, s (reactant → product)
22 stored images per path